# The `exonot` package: standardized exoplanet notation

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**Abstract**

We present transmission spectroscopy of the hot Saturn WASP-39 b, a $0.28\,M_{\mathrm{J}}$ planet orbiting its host $A_\star$ with a period $P_{\mathrm{orb}} = 4.06\,\mathrm{d}$ at a semi-major axis $a = 0.0486\,\mathrm{au}$. The retrieved spectrum shows strong $H_2O$, $CO_2$, and $SO_2$ absorption, with tentative CO and $CH_4$, in a $H_2$/He dominated atmosphere of mean molecular weight $\mu \approx 2.3$ and carbon-to-oxygen ratio C/O $\approx 0.35$. The inferred equilibrium temperature is $T_{\mathrm{eq}} \approx 1170\,\mathrm{K}$ and the atmospheric scale height $H$ is large enough to yield a transit signal of several hundred ppm, detected at S/N $> 20$ per resolution element.

## 1 System and stellar parameters

The host star $A_\star$ has effective temperature $T_{\mathrm{eff}}$, surface gravity $\log g$, luminosity $L_\odot$, mass $M_\star = 0.93\,M_\odot$, radius $R_\star = 0.90\,R_\odot$, and metallicity [Fe/H] $= -0.10$. The orbit is consistent with circular ($e < 0.05$) and well aligned ($\lambda \approx 0^\circ$), with inclination $i = 87.8^\circ$ and radial-velocity semi-amplitude $K = 33\,\mathrm{m\,s^{-1}}$. Planetary parameters for WASP-39 b are collected in Table ??. Companion candidates would be labelled c, d, and so on.

Table 1: Measured and derived parameters of WASP-39 b.

| Parameter | Symbol | Value |
|---|---|---|
| Planet mass | $M_{\mathrm{p}}$ | $0.28\,M_{\mathrm{J}}$ |
| Planet radius | $R_{\mathrm{p}}$ | $1.27\,R_{\mathrm{J}}$ |
| Planet density | $\rho_{\mathrm{p}}$ | $0.17\,\mathrm{g\,cm^{-3}}$ |
| Orbital period | $P_{\mathrm{orb}}$ | $4.06\,\mathrm{d}$ |
| Semi-major axis | $a$ | $0.0486\,\mathrm{au}$ |
| Eccentricity | $e$ | $< 0.05$ |
| RV semi-amplitude | $K$ | $33\,\mathrm{m\,s^{-1}}$ |
| Equilibrium temperature | $T_{\mathrm{eq}}$ | $1170\,\mathrm{K}$ |
| Surface gravity | $\log g$ | 2.63 |
| Impact parameter | $b$ | 0.45 |
| Transit depth | $(R_{\mathrm{p}}/R_\star)^2$ | 0.0211 |
| Total transit duration | $T_{14}$ | $2.80\,\mathrm{h}$ |
| Ingress/egress duration | $T_{12}$ | $0.24\,\mathrm{h}$ |
| Limb-darkening coeff. | $u_1$ | 0.42 |
| Mean molecular weight | $\mu$ | 2.3 |

## 2 Atmospheric scale height

The pressure scale height that sets the amplitude of the transmission signal follows from hydrostatic equilibrium,

$$H = \frac{k_{\mathrm{B}}\,T_{\mathrm{eq}}}{\mu\,m_{\mathrm{H}}\,g}, \tag{1}$$

where $k_B$ is the Boltzmann constant, $m_H$ the mass of a hydrogen atom, and $g$ the surface gravity (here written as $\log g$). The corresponding transit depth is

$$\delta = (R_p/R_\star)^2 , \tag{2}$$

so the spectral modulation across the transmission spectrum $R_\lambda$ scales as $2H/R_\star$ in units of $(R_p/R_\star)^2$.

# 3 Composition

Retrieved volume mixing ratios for the dominant absorbers are $X_{H_2O} \sim 10^{-3}$, $X_{CO_2} \sim 10^{-4}$, $X_{CO} \sim 10^{-4}$, and $X_{CH_4} < 10^{-6}$, alongside a photochemically produced $SO_2$ component and high-altitude hazes. Optical absorbers such as Na I, K I, TiO, and VO are not required by the data, nor are $H_2O$-dissociation tracers like $H^-$ or $e^-$. Model comparison favours the cloudy, $SO_2$-bearing fit with $\Delta\mathrm{BIC} \approx -32$ (equivalently a Bayes factor $\ln \mathcal{B} \approx 16$ and $\chi^2_\nu \approx 1.05$). After correcting for a stellar contamination factor $f_{cont} \approx 0.99$, the data remain consistent with a $H_2$/He envelope of metallicity [Fe/H] $\approx +1$. Earth ($\oplus$) and Sun ($\odot$) symbols are available when the `symbols` option is loaded.